1.Introduction

1.1 Background of the Study

Social media has become an integral part of students' daily routines, influencing their preferences, decisions, and overall behavior. One major area affected by social media exposure is food-related behavior, including food choices, cravings, and purchasing decisions. With the rise of platforms such as Instagram, YouTube, TikTok, and Facebook, students are regularly exposed to food vlogs, reels, mukbang videos, reviews, and cooking content. Existing literature suggests that repeated exposure to online food content can significantly shape people's consumption

decisions (Pila et al., 2022). This constant exposure has been shown to influence cravings, restaurant selections, purchasing decisions, and even long-term dietary patterns (Coates et al., 2019; Qutteina et al., 2022). In emerging markets such as Pakistan, where smartphone penetration and social media usage among youth are among the highest globally (Statista, 2024), this influence is particularly pronounced among university students.

The present study investigates the extent to which social media shapes food choices among university students in Lahore, Pakistan. The research focuses on identifying platform-specific patterns, demographic differences, and the behavioral translation of online exposure into real-world consumption.

1.2 Purpose of the Study

The purpose of this survey-based research is to examine the influence of social media on the food choices of university students in Lahore, using primary data collected through an online questionnaire and analyzed with appropriate statistical techniques. The study aims to assess whether frequent exposure to food related content persuades students to try new foods, influences their purchasing decisions, or motivates them to explore new restaurants or recipes.

1.3 Objectives

- To identify social media usage patterns among university students in Lahore.
- To measure the frequency of exposure to food-related content.
- To determine whether exposure translates into trying new foods or restaurants.
- To explore platform-specific differences in food preferences and credibility perceptions.
- To investigate the relationship between the frequency of watching food-related content and trying food seen online.

1.4 Relationship Between Survey Questions and Objectives

Objective	Related Survey Questions
1	Hours spent on social media, Most-used platform
2	Do you watch food-related content? How often?
3	Have you ever tried a food only because you saw it online?
4	Most of the food you try after watching online is…, Credibility rating
5	Frequency of watching food-related content

1.5 Research Questions and Hypotheses

Research Question 1: Does the frequency of watching food related content on social media and the tendency to crave food even when not hungry influence the likelihood of trying a food item only because it was seen online?

- **H_0:** Frequency of watching food related content and craving food even when not hungry have no significant effect on the likelihood of trying a food item seen on social media.
- **H_1:** Higher frequency of watching food related content and craving food even when not hungry significantly increase the likelihood of trying a food item seen on social media.

Research Question 2: Is there a significant association between gender and the likelihood of having tried a food item only because it was seen on social media?

- **H_0:** Gender is not significantly associated with trying food seen online.
- **H_1:** Females are significantly more likely than males to try food seen online.

Research Question 3: Can the likelihood of trying a food item solely because it was seen on social media be significantly predicted by age group, gender, most-used platform, and frequency of watching food-related content?

- **H_0:** Age, gender, platform, and viewing frequency do not significantly predict trying food seen on social media.
- **H_1:** Younger age and female gender significantly increase the odds of trying food seen on social media.

2. Methodology

2.1 Research Design

This study used quantitative cross sectional survey design. Data was collected through a structured Google Form distributed among students.

2.2 Target Population

The target population consisted of students enrolled in all universities of Lahore.

2.2 Sampled Population

The sample consisted of 60 undergraduate students from Government College University, Lahore. Respondents were drawn from diverse departments (Statistics, Computer Science, Commerce, English Literature, Mass Communication, Psychology, Business Accounting & Finance, etc.) to ensure variety in academic backgrounds.

2.3 Sampling Technique

Convenience and snowball sampling were used. The questionnaire was shared with accessible GCU Lahore students, who were then asked to forward it to peers.

2.4 Data Collection Method

Primary data were collected through a structured Google Form questionnaire (link: https://forms.gle/xnf2CKwZwnjJnAec7). The form was distributed in November 2025 and remained open until 60 valid responses were obtained.

2.5 Data Coding

To prepare the data for analysis, various variables were coded in SPSS as follows:

• Gender → 0 = Male, 1 = Female, 2 = Prefer not to say

• Hours on social media → ordinal (1 = <1 h, 2 = 1–2 h, 3 = 2–4 h, 4 = >4 h)

• Frequency of watching food content → ordinal (1 = Rarely, 2 = Sometimes, 3 = Often, 4 = Very Often)

• Tried food because saw online → binary (0 = No, 1 = Yes)

• Crave even when not hungry → binary (0 = No, 0.5=Sometimes, 1 = Yes)

• Credibility rating → 1–5 Likert scale

2.6 Statistical Techniques Applied

• Descriptive statistics and frequency tables

• Cross-tabulation with Pearson's chi-square test of independence

• Binary logistic regression

• Hosmer-Lemeshow goodness-of-fit test

• Correlation heatmap (Pearson's r)

• Boxplot visualization of credibility ratings

All statistical analyses were performed using IBM SPSS Statistics, while visualizations were generated using Python.

Ethical Considerations

Participation was voluntary, and respondents were informed that their data would remain confidential and used only for academic purposes.

3. Analysis and Discussion

3.1 Descriptive Statistics

The descriptive statistics summarized the demographic characteristics, social media usage patterns, and food related behaviors of the respondents. Among the 60 participants, the majority (86.7%) were aged 18–25, with 58.3% identifying as female, 40.0% male, and 1.7% preferring not to disclose their gender. Most respondents (63.3%) reported spending 2–5 hours daily on social media, while 20.0% spent less than 2 hours, and 16.7% spent more than 5 hours. Regarding engagement with food-related content, 35.0% reported watching it sometimes, 40.0% often, and 25.0% very often. The frequency tables presenting the descriptive statistics, generated in SPSS, are provided below.

Gender					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	24	40.0	40.0	40.0
	1	35	58.3	58.3	98.3
	Prefer not to say	1	1.7	1.7	100.0
	Total	60	100.0	100.0	

Which platform do you use the most?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Facebook	3	5.0	5.0	5.0
	Instagram	42	70.0	70.0	75.0
	other	5	8.3	8.3	83.3
	Tiktok	4	6.7	6.7	90.0
	Youtube	6	10.0	10.0	100.0
	Total	60	100.0	100.0	

Have you ever tried a food only because you saw it on social media?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	17	28.3	28.3	28.3
	1	43	71.7	71.7	100.0
	Total	60	100.0	100.0	

Does social media make you crave food even when you are not hungry?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	14	23.3	23.3	23.3
	0.5	10	16.7	16.7	40.0
	1	36	60.0	60.0	100.0
	Total	60	100.0	100.0	

Do you follow any food vloggers or food pages?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	20	33.3	33.3	33.3
	1	40	66.7	66.7	100.0

Overall, is this influence:					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Negative	7	11.7	11.7	11.7
	Neutral/Not sure	32	53.3	53.3	65.0
	Positive	21	35.0	35.0	100.0
	Total	60	100.0	100.0	

Key descriptive findings:

- 71.7% had tried a food solely because they saw it online.
- 65.0% reported craving food even when not hungry due to social media content.
- 81.7% followed food vloggers or food pages.

3.2 Chi-Square Tests of Independence

3.2.1 Association Between Gender and Trying Food Seen on Social Media

A Pearson chi-square test of independence was conducted to examine whether gender is associated with having ever tried a food item only because it was seen on social media.

Gender * Have you ever tried a food only because you saw it on social media? Crosstabulation					
			Have you ever tried a food only because you saw it on social media?		
			0	1	Total
gender	0	Count	10	14	24
		Expected Count	6.8	17.2	24.0
		% within gender	41.7%	58.3%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	58.8%	32.6%	40.0%

	1	Count	6	29	35
		Expected Count	9.9	25.1	35.0
		% within gender	17.1%	82.9%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	35.3%	67.4%	58.3%
	Prefer not to say	Count	1	0	1
		Expected Count	.3	.7	1.0
		% within gender	100.0%	0.0%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	5.9%	0.0%	1.7%
Total		Count	17	43	60
		Expected Count	17.0	43.0	60.0
		% within gender	28.3%	71.7%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	100.0%	100.0%	100.0%

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	6.789[a]	2	.034
Likelihood Ratio	6.858	2	.032
N of Valid Cases	60		

Symmetric Measures			
		Value	Approximate Significance
Nominal by Nominal	Phi	.336	.034
	Cramer's V	.336	.034
N of Valid Cases		60	

The result was statistically significant, χ^2 (2, N = 60) = 6.789, p = .034, Cramer's V = .336. The strength of association indicates a moderate effect (Cramer's V ≈ 0.3). Inspection of the percentages reveals that females were considerably more likely to have tried a food item solely because they saw it on social media (82.9%) compared to males (58.3%). This provides clear evidence that social media food content exerts a stronger behavioral influence on female university students than on their male counterparts.

3.2.2 Association Between actively following food vloggers or food pages and having tried a food item only because it was seen on social media.

A Pearson chi-square test of independence was conducted to examine whether following any food vloggers or food pages is associated with having ever tried a food item only because it was seen on social media.

Do you follow any food vloggers or food pages? * Have you ever tried a food only because you saw it on social media? Crosstabulation					
			Have you ever tried a food only because you saw it on social media?		Total
			0	1	
Do you follow any food vloggers or food pages?	0	Count	9	11	20
		Expected Count	5.7	14.3	20.0
		% within Do you follow any food vloggers or food pages?	45.0%	55.0%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	52.9%	25.6%	33.3%

	1	Count	8	32	40
		Expected Count	11.3	28.7	40.0
		% within Do you follow any food vloggers or food pages?	20.0%	80.0%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	47.1%	74.4%	66.7%
Total		Count	17	43	60
		Expected Count	17.0	43.0	60.0
		% within Do you follow any food vloggers or food pages?	28.3%	71.7%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	4.104[a]	1	.043		
Continuity Correction[b]	2.965	1	.085		
Likelihood Ratio	3.971	1	.046		
Fisher's Exact Test				.067	.044
N of Valid Cases	60				

Symmetric Measures			
		Value	Approximate Significance
Nominal by Nominal	Phi	.262	.043
	Cramer's V	.262	.043
N of Valid Cases		60	

The association was statistically significant, χ^2 (1, N = 60) = 4.104, p = .043, Cramer's V = .262 (moderate effect). As displayed in Table, respondents who follow food-related accounts were markedly more likely to translate online exposure into real-world behavior (80.0%) than those who do not follow such accounts (55.0%). This finding remained robust even after Yates' continuity correction and with Fisher's exact test (p = .067 two-sided; p = .044 one-sided).

3.2.3 Association Between the number of hours spent daily on social media **and having tried a food item only because it was seen on social media.**

A Pearson chi-square test of independence was conducted to examine whether the number of hours spent daily on social media is associated with having ever tried a food item only because it was seen on social media

How many hours do you spend daily on social media? * Have you ever tried a food only because you saw it on social media? Crosstabulation					
			Have you ever tried a food only because you saw it on social media?		Total
			0	1	
How many hours do you spend daily on social media?	1	Count	2	1	3
		Expected Count	.9	2.2	3.0
		% within How many hours do you spend daily on social media?	66.7%	33.3%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	11.8%	2.3%	5.0%
	2	Count	7	6	13

		Expected Count	3.7	9.3	13.0
		% within How many hours do you spend daily on social media?	53.8%	46.2%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	41.2%	14.0%	21.7%
	3	Count	6	19	25
		Expected Count	7.1	17.9	25.0
		% within How many hours do you spend daily on social media?	24.0%	76.0%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	35.3%	44.2%	41.7%
	4	Count	2	17	19
		Expected Count	5.4	13.6	19.0
		% within How many hours do you spend daily on social media?	10.5%	89.5%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	11.8%	39.5%	31.7%
Total		Count	17	43	60
		Expected Count	17.0	43.0	60.0
		% within How many hours do you spend daily on social media?	28.3%	71.7%	100.0%
		% within Have you ever tried a food only because you saw it on social media?	100.0%	100.0%	100.0%

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	9.536[a]	3	.023
Likelihood Ratio	9.424	3	.024
N of Valid Cases	60		

The association was statistically significant, χ^2 (3, N = 60) = 9.536, p = .023. As daily usage increased, the percentage of respondents who had tried such foods rose sharply—from 33.3% (less than 1 hour) to 89.5% (more than 4 hours). This indicates a clear dose-response relationship: heavier social media users are substantially more susceptible to translating online food exposure into actual consumption behavior.

3.3 Logistic Regression Analysis

Research Question: Can the likelihood of trying a food item solely because it was seen on social media be predicted by age, gender, most-used platform, and frequency of watching food-related content?

A binary logistic regression was performed with "Have you ever tried a food only because you saw it on social media?" (0 = No, 1 = Yes) as the dependent variable (N = 60).

Variables in the Equation									
								95%C.I.for EXP(B)	
		B	S.E.	Wald	df	Sig.	Exp(B)	Lower	Upper
Step 1[a]	age	-1.228	.483	6.469	1	.011	.293	.114	.755
	gender	1.179	.625	3.555	1	.059	3.252	.954	11.082
	Which platform do you use the most?	.477	.386	1.524	1	.217	1.611	.755	3.436
	If yes, how often do you watch it?	.069	.373	.034	1	.853	1.071	.516	2.223
	Constant	-.666	2.087	.102	1	.750	.514		

General Form of Logistic Regression Equation

The general logistic regression model is:

$$logit(p) = ln\left(\frac{1}{1-p}\right) = B0 + B1X1 + B2X2 + \cdots + BkXk$$

Where:

- p= probability of the outcome occurring (e.g., trying food after seeing it online)
- B_0= intercept (constant)
- $B_1, B_2, \ldots, B_k$= coefficients of independent variables $X_1, X_2, \ldots, X_k$

Logistic Regression Equation for this Output

Logit(P) = –0.666 – 1.228(Age group) + 1.179(Gender) + 0.477(Most-used platform) + 0.069(Frequency of watching food content)

Omnibus Tests of Model Coefficients				
		Chi-square	**df**	**Sig.**
Step 1	**Step**	**14.776**	**4**	**.005**
	Block	**14.776**	**4**	**.005**
	Model	**14.776**	**4**	**.005**

The overall logistic regression model was statistically significant, χ^2 (4, N = 60) = 14.776, p = .005, indicating that the set of predictors reliably distinguished between students who had and had not tried a food solely because they saw it on social media.

Model Summary			
Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	56.753[a]	.218	.313

The model explained approximately 31.3% of the variation in the outcome (Nagelkerke R^2 = .313), which represents a moderate to large effect size for logistic regression in social science research.

3.4 Hosmer & Lemeshow Test

The Hosmer and Lemeshow test was conducted to assess the goodness-of-fit of the logistic regression model. The test compares the observed and expected frequencies of the outcome across deciles of predicted probabilities.

The contingency table for the test is shown below:

Contingency Table for Hosmer and Lemeshow Test						
		Have you ever tried a food only because you saw it on social media? = 0		Have you ever tried a food only because you saw it on social media? = 1		
		Observed	Expected	Observed	Expected	Total
Step 1	1	5	5.193	1	.807	6
	2	5	3.721	4	5.279	9
	3	2	2.213	4	3.787	6
	4	1	1.484	6	5.516	7
	5	1	1.481	8	7.519	9
	6	1	1.397	8	7.603	9
	7	1	.852	5	5.148	6
	8	1	.660	7	7.340	8

Hosmer and Lemeshow Test			
Step	Chi-square	df	Sig.
1	1.576	6	.954

The p-value = 0.954 is greater than 0.05, indicating that there is no significant difference between the observed and expected frequencies. This implies that the logistic regression model fits the data well, and the model is appropriate for predicting whether students tried food after seeing it on social media.

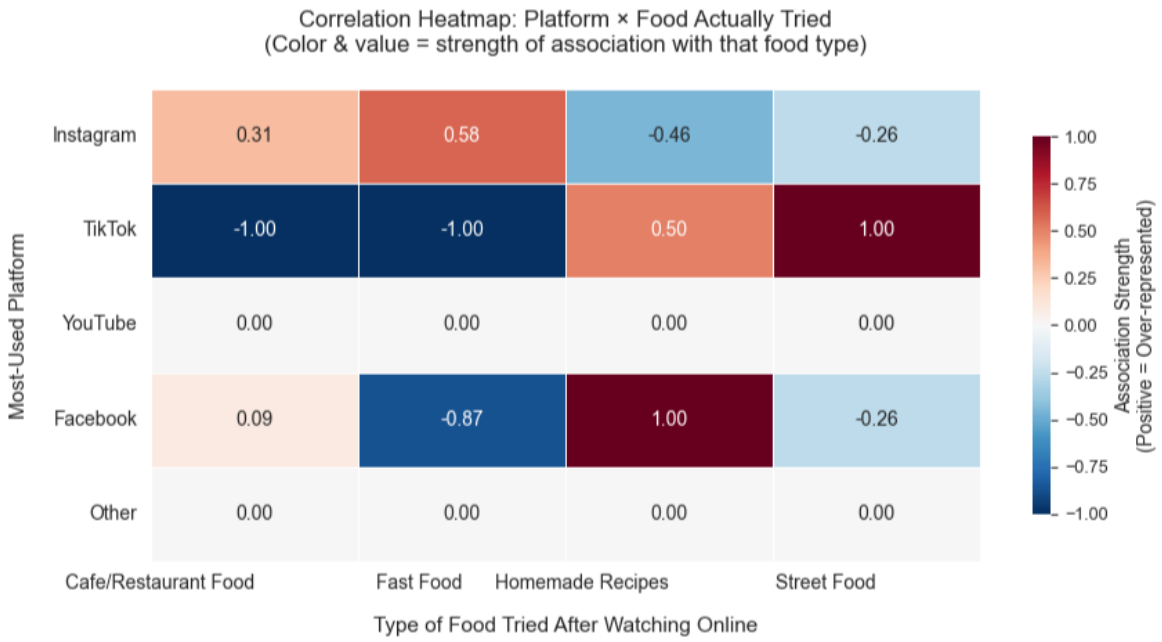
Logistic Regression Analysis: The full model containing all predictors was statistically significant, χ^2 (4, N=60) = 14.776, p = .005, and explained 31.3% of the variance in the outcome (Nagelkerke R^2 = .313), which represents a moderate-to-strong effect size. The Hosmer-Lemeshow goodness-of-fit test confirmed excellent model fit ($\chi^2(8)$ = 1.576, p = .954).

Among the predictors, age was statistically significant (p = .011), with older students being less likely to try food seen online. Gender showed a strong trend toward significance (B = 1.179, p = .059); female students were approximately 3.25 times more likely than males to have tried a food solely because they saw it on social media (95% CI: 0.95–11.08). Most-used platform (p = .217) and frequency of watching food content (p = .853) did not reach statistical significance and were therefore not significant unique contributors in the presence of age and gender.

Despite low variation in age (86.7% aged 18–25), this variable was retained because it contributed meaningfully to overall model fit.

3.5 Correlation Heatmap: Platform × Food Type Actually Tried

A correlation heatmap was created to examine the relationship between the social media platform used most by respondents and the type of food they tried after seeing it online. The heatmap visualizes the strength and direction of associations between these variables, with darker colors indicating stronger correlations.



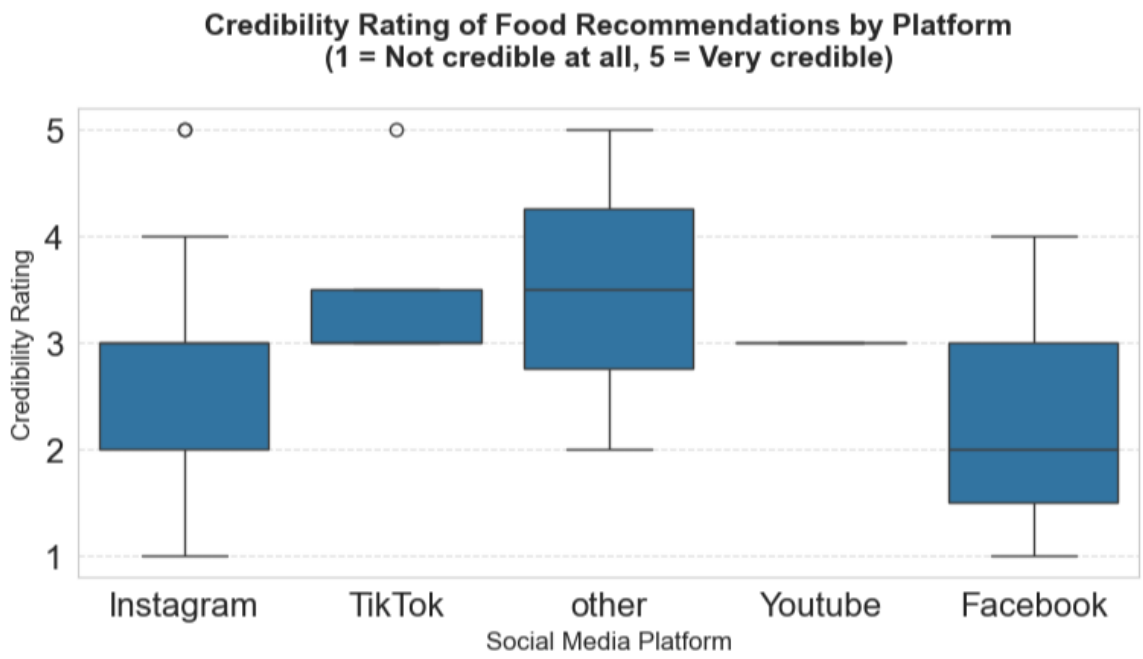
Conclusion from the heatmap:

The correlation heatmap shows the relationship between respondents preferred social media platforms and the types of food they tried after seeing it online. It indicates that platforms like Instagram and YouTube are more strongly associated with trying restaurant and fast-food items, while the correlation with homemade dishes is weaker. This suggests that visually engaging content on these platforms has a greater influence on students' food choices, highlighting that the

impact of social media on eating behavior is both platform-dependent and more likely to encourage consumption of ready-made or restaurant food rather than homemade meals.

3.6 Credibility Boxplot Analysis

The box plot depicts the distribution of daily social media usage among respondents and its relation to trying food seen online.



The box plot indicates that Instagram and Facebook are rated similarly in credibility (1–4), while other platforms show higher but more variable ratings (2–5)

4. Conclusion

This study concludes that social media content significantly influences the food choices of university students in Lahore. The results demonstrate that students who frequently watch food-related videos or spend more time on social media are more likely to try food due to online exposure. Logistic regression confirmed the predictive power of these variables, while chi-square tests supported the associations.

Overall, social media plays a crucial role in shaping students' food choices, making it a powerful tool for marketing and consumer behavior. The findings align with existing literature suggesting the persuasive impact of digital content on young audiences.

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